



Indira Gandhi National Open University New Delhi

A SYNOPSIS ON

Supermarket Electronic Management System

MCSP-232

Submitted to Dehli 1 (COE)

In the partial fulfillment of the requirements for the degree of
Master of Computer Application (MCAOL)

Project Guide

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Phd

Submitted by:

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MCAOL

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Supermarket Electronic Management System
A Comprehensive Web-Based Solution



SCHOOL OF COMPUTER AND INFORMATION SCIENCES
IGNOU, MAIDAN GARHI, NEW DELHI - 110 068

II. PROFORMA FOR THE APPROVAL OF MCA PROJECT PROPOSAL (MCSP-232)

(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

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- Name and Address of the Student: Khost Afghanistan
- Title of the Project***: Supermarket Electronic Management System
- Name and Address of the Guide: Dr. Shekhar Kalakoti
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- Educational Qualification of the Guide:
(Attach bio-data also)
(*in Computer Science / IT only)
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- Working / Teaching experience of the Guide** : 10 years

(**Note: At any given point of time, a guide should not provide guidance for more than 5 MCA students of IGNOU)

- Software used in the Project*** : Python, PostgresSQL, Javascript, CSS, HTML5, Bootstrap
(*** Please refer to section VIII of these guidelines)

- Is this your first submission?

☒ Yes

☐ No

Signature of the Student
Date:

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.....

☐☐

Approved Not Approved

Signature, Designation, Stamp of the
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Date:

Suggestions for reformulating the Project:

X. CERTIFICATE OF ORIGINALITY

This is to certify that the project report titled SuperMarket 2Electronic Man" system submitted to Indira Gandhi National Open University in partial fulfilment of the requirement for the award of the degree of MASTER OF COMPUTER APPLICATIONS, is an authentic and original work carried out by Mr. / Ms. Mohammad Nabi Hemat with Enrolment No. 2353 878752 under my guidance.

The matter embodied in this project is genuine work done by the student and has not been submitted whether to this University or to any other University / Institute for the fulfilment of the requirements of any course of study.



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Date: 29/12/2024

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PROFESSIONAL SUMMARY

Assistant Professor with 10 years academic experience. Expertise in wireless sensor networks, remote sensing, data structures, algorithms, operating systems. Ph.D. pursuing in Computer Science. Published researcher with 6 peer-reviewed journal articles and 2 book chapters. ISO 9001 certified institution experience.

CORE COMPETENCIES

Technical: Wireless Sensor Networks | Remote Sensing | Algorithm Design | Data Structures | Operating Systems | Discrete Mathematics | Theory of Computation | LIDAR Data Processing | SAR Technology | Cyber Security

Teaching & Research: UG/PG Course Instruction | Research Supervision | Curriculum Development | Paper Publishing | Academic Mentoring | Department Administration

Tools & Certifications: AI/ML Basics | Data Science Fundamentals | Artificial Intelligence | GIS Tools | LIDAR Processing | Geodata Analysis

PROFESSIONAL EXPERIENCE

Assistant Professor – Computer Science

M. B. Govt. P. G. College, Haldwani, Nainital | July 2019 – Present

- Teach undergraduate and postgraduate courses in Computer Science
- Supervise student research projects and thesis work
- Department Head administration and policy implementation
- Curriculum development aligned with NEP 2020 standards
- Member: IQAC, NAAC, Research Committee
- Nodal Officer: Samarth Program
- Coordinate IGNOU examinations and student affairs

Assistant Professor – Computer Science

M. B. Govt. P. G. College, Haldwani | 2016 – 2019

- Taught core Computer Science courses to UG/PG students
- Curriculum development and course content design
- Student mentoring and examination supervision
- Academic planning and program development

Assistant Professor – Computer Science

Graphic Era Hill University, Dehradun | 2015 – 2016

- Instruction of undergraduate and postgraduate courses
- Student mentoring and guidance
- Course assessment and evaluation

EDUCATION

Degree	Institution	Status/Grade
Ph.D. Computer Science	Kumaun University	Pursuing
M.Tech Computer Science	G B Pant Engineering College, Ghurdauri	80.5%
B.Tech Computer Science	Graphic Era University, Dehradun	74.8%
Intermediate	G I C Lohaghat	71.2%
High School	V V M Lohaghat	73.6%

PUBLICATIONS & RESEARCH

Journal Articles (Peer-Reviewed)

1. Kumar, S., & Chaudhary, K. (2024). Shortest Distance On-demand Stable Election Protocol. *Wireless Personal Communications*, 139(3), 1623-1635.
2. Kumar, S., & Chaudhary, K. (2022). Shortest Distance On-demand Stable Election Protocol. *Wireless Personal Communications*.
3. Kumar, S., Verma, S. K., & Kumar, A. (2015). Enhanced threshold sensitive stable election protocol for heterogeneous wireless sensor network. *Wireless Personal Communications*, 85(4), 2643-2656.
4. Kumar, S., & Verma, S. K. (2015). Enhanced threshold sensitive stable election protocol. *Journal of Computer Engineering*, 17(3), 27-33.
5. Sah, P., & Kumar, S. (2014). A New Approach to Distributed Energy-Efficient Cluster for Heterogeneous WSNs. *International Journal of Computer Applications*, 975, 8887.
6. Kumar, S., & Sah, P. (2015). A survey on energy efficient protocols for wireless sensor network. *International Journal of Computer Applications*, 120(2).

Book Chapters

1. Shekhar Kumar & Gaurav Joshi. Crop Monitoring System. In *Exploring the Himalayas: Land and the People*. Aadharshila Prakashan. ISBN: 978-93-81913-60-4
2. Gaurav Joshi & Shekhar Kumar. Role of Technology in Indian Culture. ISBN: 978-81-926605-7-8

Research Review Contribution

- Reviewer for Wireless Personal Communications (Springer Journal)
- Article under review in Wireless Personal Communications

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

2024

- AI/ML for Geodata Analysis (A+ Grade) – IIRS Dehradun
- LIDAR Data Processing and Application (A+ Grade) – IIRS Dehradun
- Introduction to Artificial Intelligence – Infosys Springboard
- Introduction to Data Science – Infosys Springboard
- Faculty Enablement Program – Infosys Chandigarh

2023-2024

- NEP 2020 Orientation & Sensitization Programme – UGC MMTC
- Refresher Course (A Grade) – UGC-HRDC Himachal Pradesh University
- SAR Technology Application in Disaster Management – ISRO/IIT Roorkee
- Implementation of NEP-2020 Professional Development (86%) – UGC Approved
- Capacity Building Workshop – Uttarakhand Science Education & Research Centre

2020-2022

- Ethical Hacking and Cyber Security – Federation of Community Radio Stations
- ICT Tools for Teaching and Learning – NIT Patna, MNIT Jaipur, IIT Guwahati
- UGC Induction Programme (A Grade) – UGC-HRDC Aligarh Muslim University

ACADEMIC LEADERSHIP & ORGANIZATION

Guest Lecturer & Resource Person

- Cyber Security Awareness – M.B. Govt. P.G. College (November 2025)
- Cyber Security Topic Expert – M.B. Govt. P.G. College (November 2025)
- Computer Application in Research (Pre-Ph.D. Coursework) – M.B. Govt. P.G. College (March 2024)
- OBS Tutorial for Teachers – College of Teacher Education
- Application of Mobile Applications in Teaching – M.B. Govt. P.G. College (March 2019)
- Development of Computer Skills – College of Teacher Education

Chief Speaker & Keynote Addresses

- Online and Digital Education - Equitable Use of Technology – D.I.E.T Udham Singh Nagar (September 2020)
- Use of ICT in Classroom Practices and Possibilities – D.I.E.T Udham Singh Nagar (February 2020)

Workshop Organization & Coordination

- Organizing Member: National Workshop Basic Concepts of Mathematics & Quantum Mechanics (January 2025)

- Organizer: National Workshop on Advances in Computer Networking (February 2020)
- Organizer: Workshop 75 Years of Independence - Girl Child Education (February 2023)
- Organizer: National Workshop on IPR (March 2021)
- Organizer: Government e-Marketplace (GeM) Workshop (January 2020)
- Organizer: Developing Synergy Between ICT & Academics (March 2017)

Workshop Participant

- 75 Years of Independence: Tribal Education and Challenges (November 2024)
- NAPS Technical Training and SDP Awareness (October 2022)
- Industry Academia Gap: Diagnosis and Remedies – College of Technology Pantnagar (February 2020)
- Intellectual Property Rights – Government Degree College Garurbanj (February 2020)

Conference & Seminar Participation

- International Seminar on Sustainable Homestay Tourism (February 2025)
- Uttarakhand Educator's Summit (December 2019)
- National Conference on Leveraging Technology for Education – MIET Kumaon (November 2019)
- National Seminar on Progression and Innovation in Higher Education (November 2019)

Expert Designation

- Subject Expert in Computer Science – EMRS Khatima Interview Board (November 2023)

ADMINISTRATIVE RESPONSIBILITIES

- **Head:** Department of Computer Science (2019 – Present)
- **Nodal Officer:** Samarth Program (Present)
- **Committee Member:** IQAC, NAAC, Research Committee
- **Assistant Coordinator:** IGNOU (2018 – 2025)

AWARDS & SCHOLARSHIPS

- MHRD Monthly Scholarship – 12,400/- per month (2 years) – Master's Degree
- Master's Degree with Honors
- First Prize Winner – National Science Exhibition

PROFESSIONAL MEMBERSHIPS

- Member, Institute of Electrical and Electronics Engineers (IEEE)
-

RESEARCH INTERESTS

Wireless Sensor Networks | Design and Analysis of Algorithms | Theory of Computation |
Data Structures | Operating Systems | Discrete Mathematics | Remote Sensing | LIDAR
Technology | Geodata Analysis | Cyber Security

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1 Synopsis Title:

This Synopsis presents the development and implementation of a comprehensive ***Supermarket Electronic Management System*** built using Django web framework. The system provides an integrated solution for managing multiple aspects of supermarket operations, including inventory management, billing, financial tracking, multi-organization support, and user management. The application is designed to handle complex business requirements such as purchase orders, sales transactions, payment processing, expense tracking, and asset management while supporting multiple organizations and locations with role-based access control.

2 Introduction:

The ***SuperMarket Electronic Management System (SMEMS)*** is designed to automate day-to-day supermarket operations including asset management, billing, products, stock control, loans, users, and organizational configuration. The system replaces traditional manual processes, improves accuracy, ensures real-time updates, and provides centralized control over supermarket activities.

This system is built for multi-branch supermarkets and supports organizational hierarchy, authentication, inventory lifecycle, financial assets, billing workflow, and reporting.

2.1 System Objectives

The primary objectives of this system are:

- **Secure access:** using to allow authorized users through login authentication.
- **Generate reports:** to generate daily, monthly and yearly reports of sales, stock and profit.
- **Mange information:** to maintain customer's records, history and loyalty information's.
- **Save time and Cost:** to save time and cost of the products.
- **Decision Making:** it is best for decision making.
- **Reduce human errors:** it will reduce the human errors.
- **Improve billing efficiency:** it is using to improve efficiency.
- **To automate Supermarket Operations:**
- **TO support decision making.**

Synopsis Title: Supermarket Electronic Management System

2.2 Scope of the System

The system encompasses the following functional modules:

- **Products & Stocks Management Module:** Products, categories, units, stock management, and product details
- **Billing & Sales Management Module:** Purchase bills, sales bills, payment processing, receivable tracking
- **Assets & Financial Accounts Management Module:** Organizational assets, financial summaries, profit/loss statements
- **Loans & Asset Summaries**
- **User Management Module:** User registration, authentication, organization membership, role assignments
- **Configuration Module:** Organizations, locations, currencies, system settings
- **Expenditure Module:** Expense tracking and categorization

The system is built as a web-based application accessible through standard web browsers, deployed using PostgreSQL database backend, and designed for both local and cloud deployment scenarios.

3. Tools Platform

Software and Technologies to be used in this Project:

3.1 Software platform

Languages to be Used Front-end:

- HTML5
- CSS3
- JavaScript
- Bootstrap
- AJAX
- Template Engine

Back End Languages:

Synopsis Title: Supermarket Electronic Management System

- Django 4.x (Python)
- PostgreSQL

API Framework

- Django REST Framework

Additional Libraries and Tools

- Jalali Date
- Django CORS Headers
- Pillow
- Whitenoise
- Gunicorn
- Django Heroku

3.2 System Requirements

- Hardware Requirements (Server)
- Processor: 2+ CPU cores
- RAM: 4GB minimum, 8GB recommended
- Storage: 50GB minimum, SSD recommended
- Network: 100Mbps bandwidth

3.3 Software Requirements (Server)

- Operating System: Linux (Ubuntu 20.04+) or Windows Server
- Python: 3.8 or higher
- PostgreSQL: 12 or higher
- Web Server: Nginx or Apache (for production)

3.4 Client Requirements

- Modern web browser (Chrome, Firefox, Safari, Edge)
- Internet connection (minimum 1Mbps)
- Screen resolution: 1024x768 minimum, 1920x1080 recommended

3.5 System Modules

The system follows the Model-View-Template (MVT) architectural pattern, which is Django's variation of the traditional MVC pattern:

- Models Layer
- Views Layer
- Templates Layer

3.6 Database Schema Design

Core Entities

- Organization
- User and Organization User
- Product
- Bill
- Stock

3.7 Relationship Structure

The relationship structure of **SMEMS** is the complete set of entity-to-entity connections that define how:

- Products relate to Categories and Stock
- Bills relate to Products and Assets
- Loans relate to Installments
- Users relate to Groups and Organizations
- Organizations relate to Country, Currency, and Location

This structure forms the **core database design** and **UML structure** of the supermarket management system.

SMEMS contains six major modules:

1. Product & Stock Module
2. Billing Module
3. Asset Management Module
4. Loan Module
5. Authentication & Authorization Module
6. Configuration Module

Each module has entities (tables/classes), and they link with each other through:

- One-to-many relationships
- Many-to-one relationships
- Many-to-many relationships
- Derived/summary relationships

3.8 Security Architecture

- Authentication
- Authorization
- Data Protection
- Session Management

4 Core Functional Modules**4.1 Product Management Module**

- Product Catalog
- Category Management
- Unit Management
- Stock Management

4.2 Bill Management Module

- Bill Types and Structure

- Purchase Bill Processing
- Sales Bill Processing
- Payment and Receivable Tracking
- Bill Approval Workflow

4.3 Asset Management Module

- Asset Tracking
- Liability Tracking
- Receivables Management
- Financial Summary Tables
- Profit and Loss Statements

4.4 User Management Module

- User Registration and Authentication
- Role-Based Access Control
- Organization User Model

4.5 Configuration Module

- Organization Management
- Location Management
- Currency and Localization

5. Technical Implementation Details

- Database Optimization Techniques
- Indexing Strategy
- Query Optimization
- Transaction Management

5.1 Business Logic Implementation

- Stock Movement Tracking

- Profit Calculation
- Asset Summary Updates

5.2 API Design and Implementation

- RESTful API Structure
- Serialization

5.3 Frontend Implementation

- Responsive Design
- Dynamic Forms
- Reporting Interface

5.4 Deployment Architecture

- Development Environment
- Production Environment
- Configuration Management

6 Testing and Quality Assurance

6.1 Testing Strategy

- Unit Testing
- Integration Testing
- Functional Testing

6.2 Performance Testing

- Load Testing
- Query Performance Analysis

6.3 Security Testing

- Vulnerability Assessment

- Penetration Testing

7 Results and Analysis

7.1 System Capabilities Achieved

- Comprehensive Inventory Management
- Financial Transaction Processing
- Multi-Organization Support
- Performance Metrics

7.2 User Acceptance and Feedback

- Usability Evaluation
- Business Impact

7.3 Limitations and Constraints

- Current Limitations
- Scalability Considerations

8. Data Flow Diagram:

8.1 What is Data Flow Diagram (DFD) ?

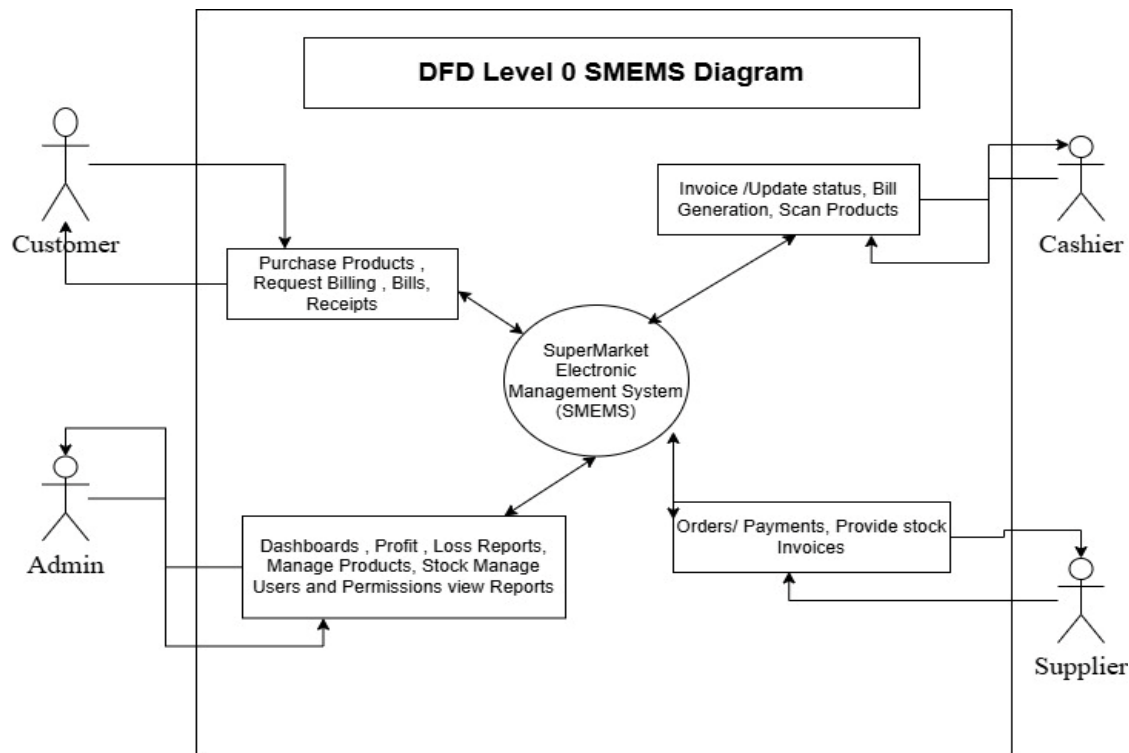
Data Flow Diagram (DFD) is a graphical representation of data flow in any system. It is capable of illustrating incoming data flow, outgoing data flow and store data. The DFD depicts both incoming and outgoing data flows and provides a high-level overview of system functionality. It is a relatively simple technique to learn and use, making it accessible for both technical and non-technical stakeholders.

Data Flow Diagram can be represented in several ways. The Data Flow Diagram (DFD) belongs to structured-analysis 1 tools. Data Flow diagrams are very popular because they help us to visualize the major steps and data involved in software-system processes.

8.2 DFD Level-0 (Context Diagram)

A Level-0 DFD (Context Diagram) for a SuperMarket Electronic Management System (SMEMS) is the highest-level data flow diagram. It shows the entire supermarket system as one single process and represents:

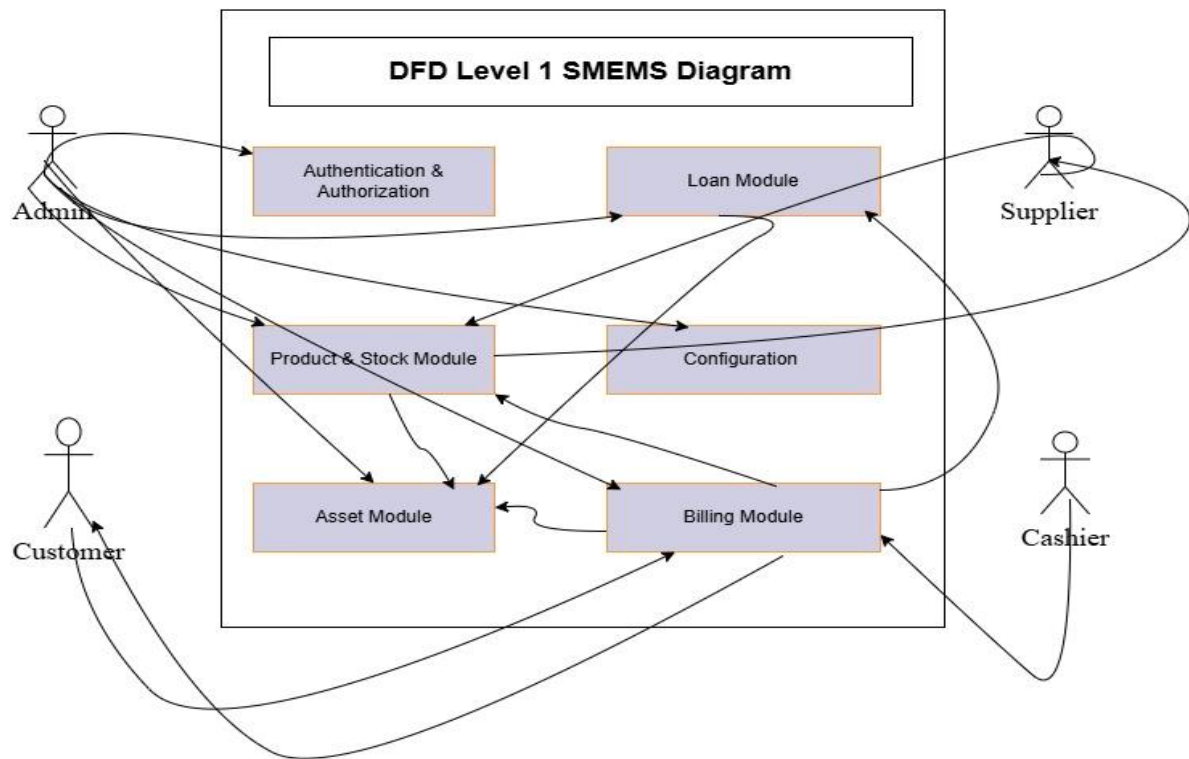
- External entities (actors)
- Major data flows between those entities and the system
- The system as one black-box process
- No internal modules shown
- DFD Level 0 (Context Diagram): showing the system as a whole and its interactions with external entities.



[Fig 1 DFD Level 0 SMEMS diagram](#)

8.3 Level-1 Data Flow Diagram (DFD)

A Level-1 DFD for a SuperMarket Electronic Management System (SMEMS) takes the single process from Level-0 and breaks it into major subsystem processes. This level shows internal modules and how data flows among them.



[Fig 2 DFD Level 1 SMEMS Diagram](#)

8.4 ERD Diagram

An ERD (Entity–Relationship Diagram) for a **SuperMarket Electronic Management System (SMEMS)** is a database modeling diagram that shows:

- Entities (tables)
- Attributes (fields)
- Relationships between entities

It describes how data is stored, linked, and structured within the supermarket management system.

The ERD helps you understand:

- How *products, stocks, bills, users, loans, assets, and configurations* are related
- How data flows across modules
- How tables connect via primary and foreign keys
- How to design the database schema

ER_Diagram For SMEMS System

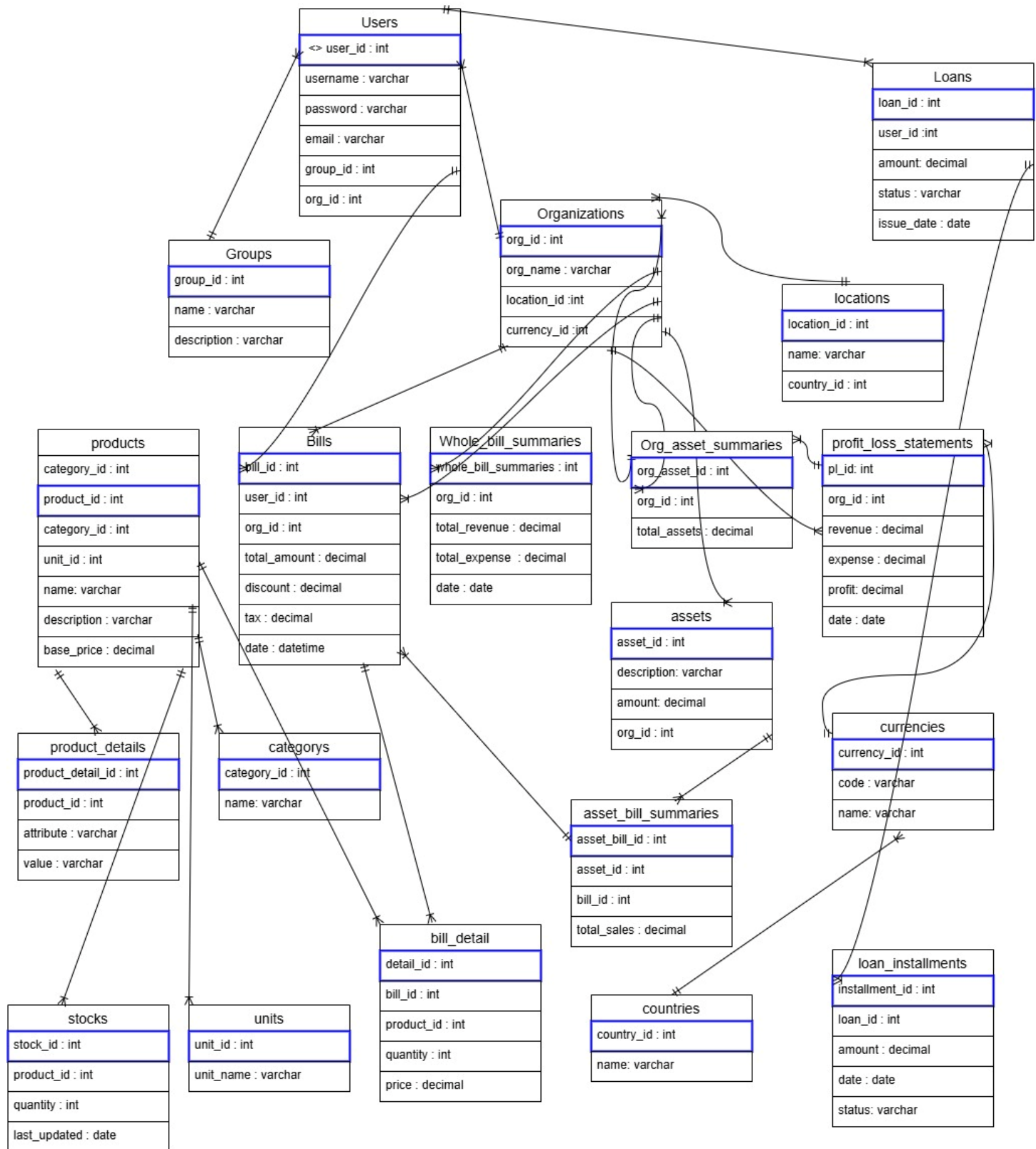


Fig 3 ER_Diagram for SMEMS System

Synopsis Title: Supermarket Electronic Management System

8.5 Use Case Diagram

A **Use Case Diagram** is a visual representation that shows how different users (actors) interact with a system to achieve various goals or “use cases.” For a **SuperMarket Electronic Management System (SMEMS)**, it depicts all the functionalities of the system and the roles of the users interacting with it.

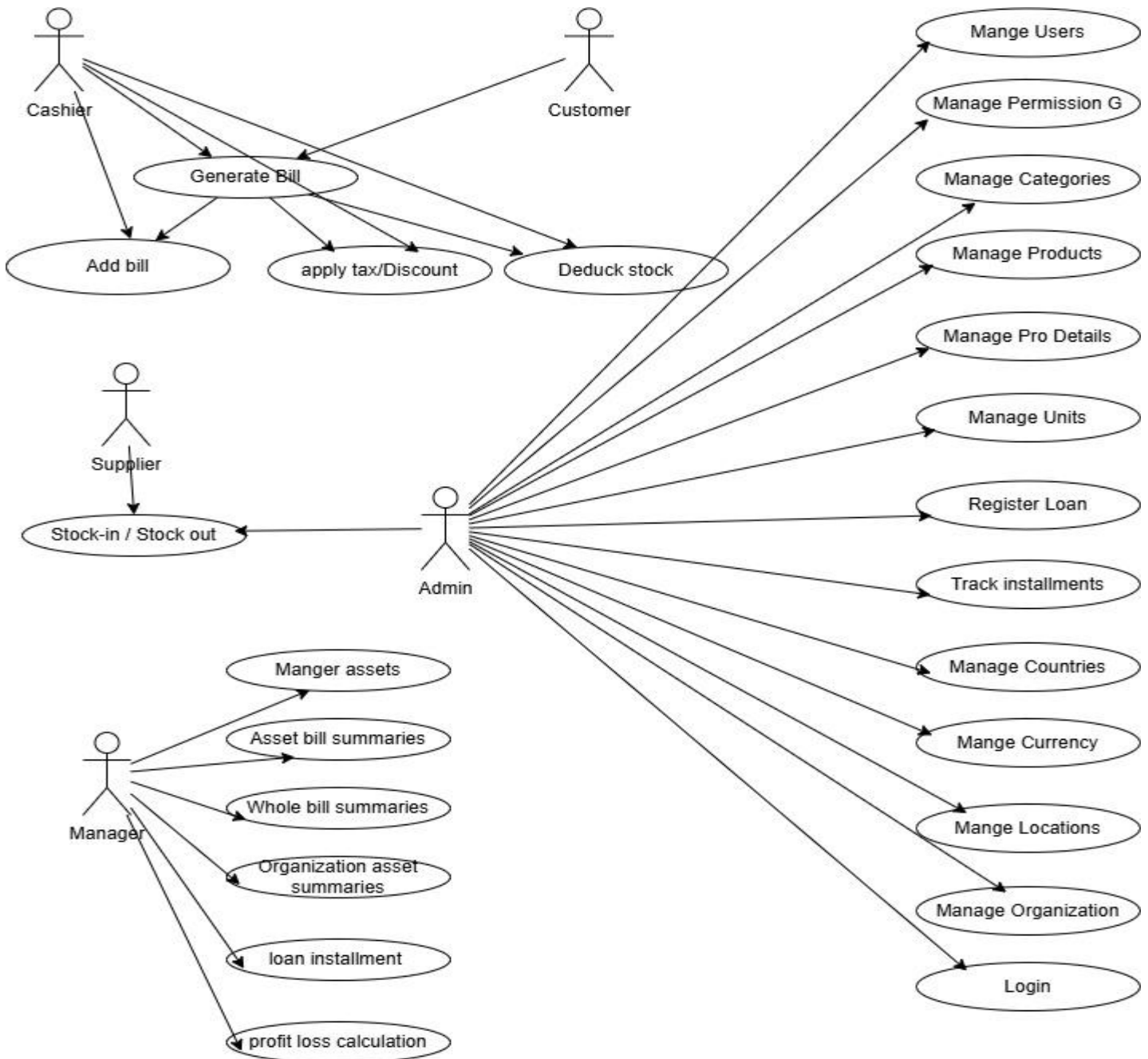


Fig 4 : Use Case diagram for SMEMS System

8.6 Activity Diagrams

An Activity Diagram for a **SuperMarket Electronic Management System (SMEMS)** shows the workflow of major processes such as billing, stock handling, asset tracking, authentication, configuration, and loan management. It visually represents how activities begin, the decisions involved, and how the workflow ends.

- Shows how processes flow in the supermarket system.
- Represents activities, decisions, parallel tasks, and end results.
- Helps understand how a user interacts with the system internally.

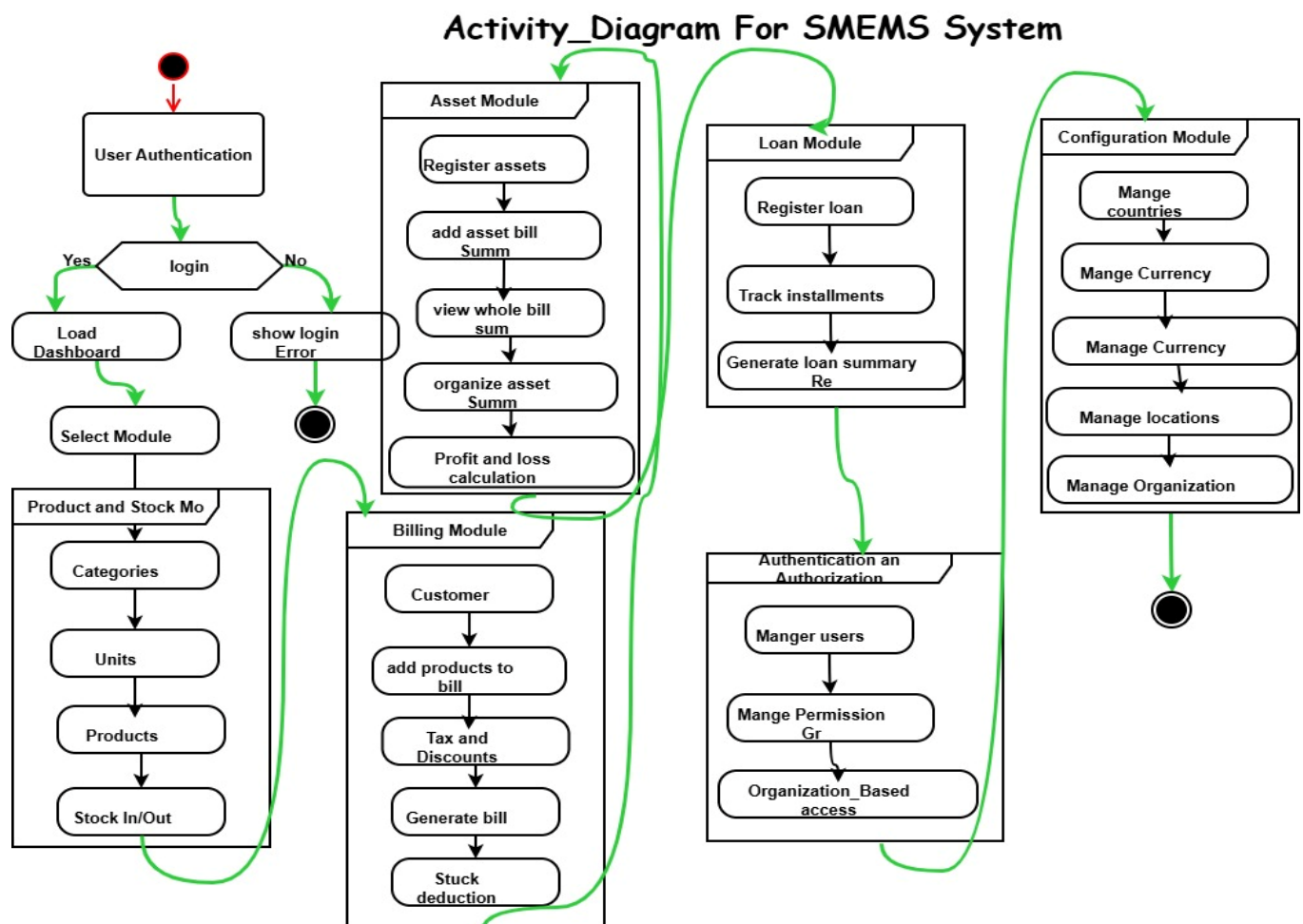


Fig 5: Activity diagram for SMEMS System

8.7 Class Diagram

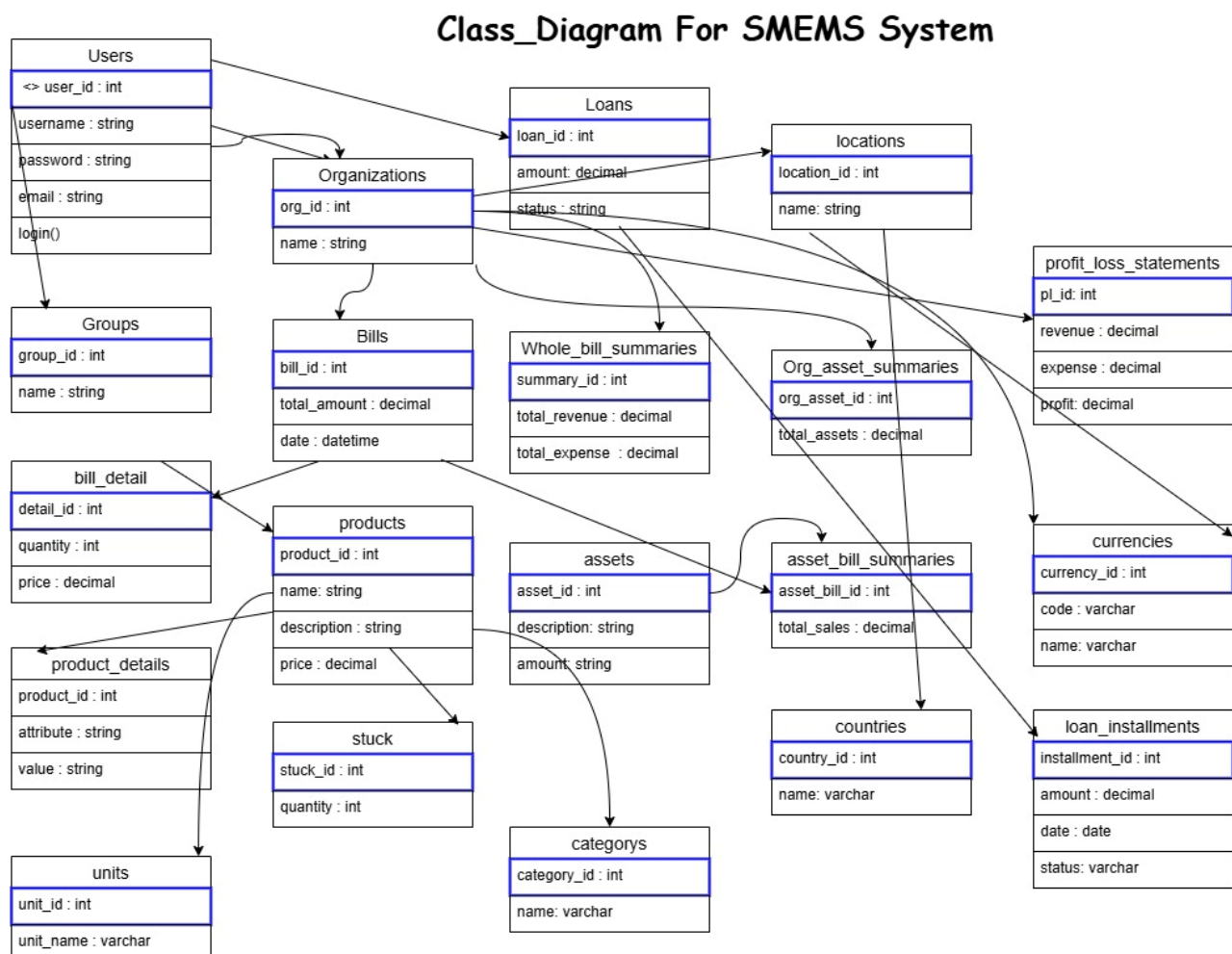
A **Class Diagram** for a *SuperMarket Electronic Management System (SMEMS)* is a structural UML diagram that shows the system's:

- Entities (classes)

Fig 6: Class diagram for SMEMS system

- Attributes
- Relationships
- Multiplicity (1..*, 0..1, etc.)
- Modules (Billing, Products, Stock, Asset, Loans, Auth, Configuration)

It is one of the most important diagrams in system design because it defines the database structure, system architecture, and application logic.



9.8 Sequence Diagram

A **Sequence Diagram** is a **UML behavioral diagram** that illustrates:

- **Actors** (User, Admin, Cashier, Supplier, etc.)
- **System components** (UI, Database, Inventory Module, Payment Module)
- **Messages exchanged** between them
- **Order of execution** from start to end

It focuses on “**who talks to whom and in what order.**”

Sequence_Diagram For SMEMS System

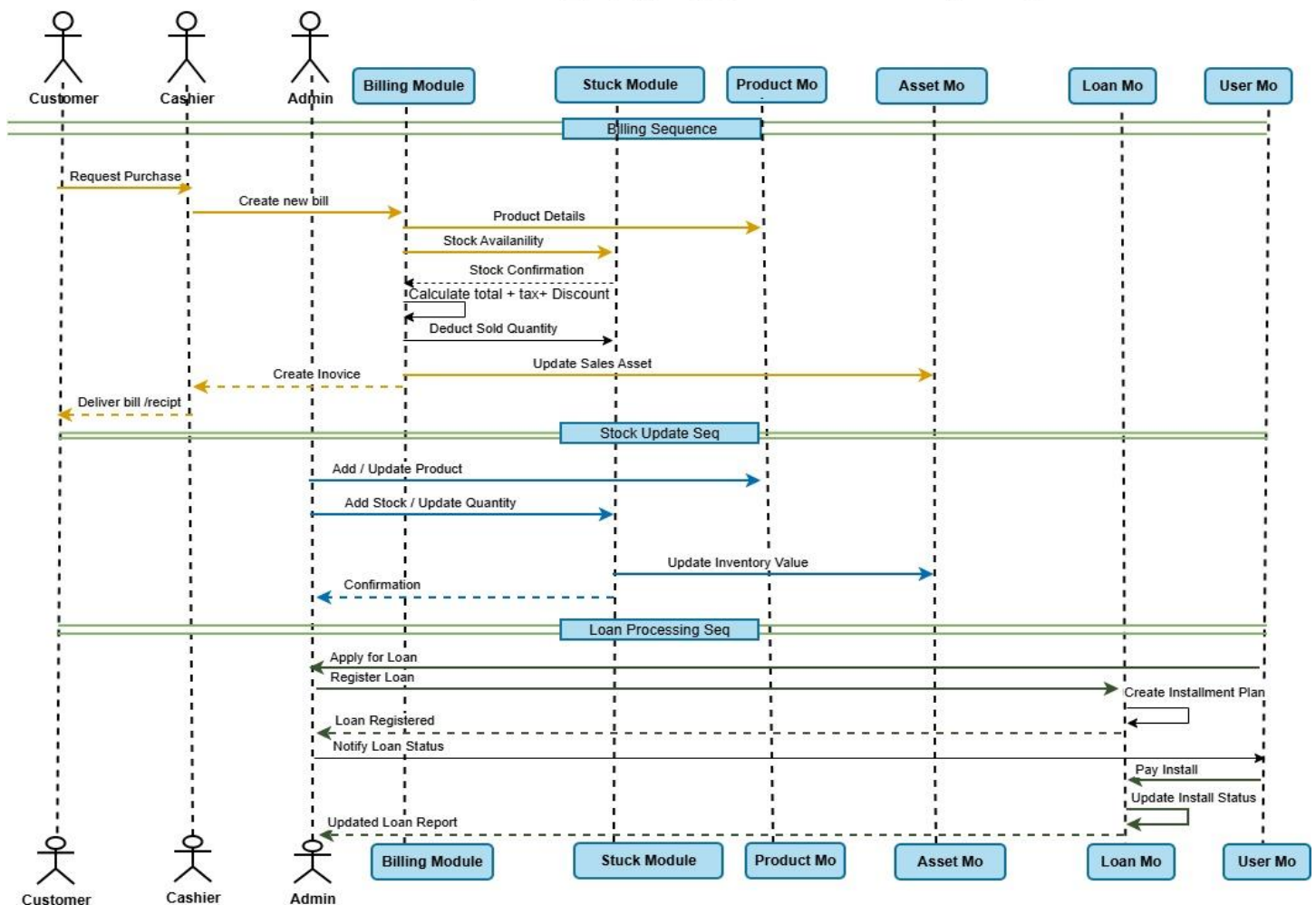


Fig 7: Sequence diagram for SMEMS System

9. Installation Guide

9.1 Local Development Setup

- Clone repository
- Create virtual environment: ``python -m venv env``
- Activate environment: ``env\Scripts\activate`` (Windows) or ``source env/bin/activate`` (Linux/Mac)
- Install dependencies: ``pip install -r requirements.txt``
- Configure database in settings.py
- Run migrations: ``python manage.py migrate``
- Create superuser: ``python manage.py createsuperuser``
- Start server: ``python manage.py runserver``

10. Production Deployment

- Configure environment variables
- Set DEBUG=False
- Configure ALLOWED_HOSTS
- Set up PostgreSQL database
- Collect static files: ``python manage.py collectstatic``
- Configure Gunicorn or Uwsgi
- Set up Nginx as reverse proxy
- Configure SSL certificates
- Set up automated backups

11. User Roles and Permissions Matrix

View Products, Create Products, Edit Products, Delete Products, Create Bills, Approve Bills, Cancel Bills, View Reports, Manage Users, System Settings, Financial Data

12. Database Schema Diagram

[Space reserved for Entity-Relationship Diagram showing all tables, relationships, and key fields]

13. API Endpoint Documentation

[Space reserved for comprehensive API documentation with request/response examples]

14. User Interface Screenshots

- [Space reserved for screenshots of key system interfaces:
- - Dashboard
- - Product Management
- - Bill Creation
- - Stock Management
- - Reports
- - User Management]

15. Sample Reports

- [Space reserved for sample report outputs:
- - Profit/Loss Statement
- - Inventory Valuation Report
- - Sales Summary Report
- - Outstanding Receivables Report
- - Purchase History Report]

16. Conclusions and Future Work

This thesis has presented the successful development and implementation of a comprehensive Supermarket Electronic Management System. The system integrates inventory management, financial tracking, multi-organization support, and user management into a unified web-based platform.

Key Achievements

Comprehensive Functionality: The system successfully handles all core supermarket operations from product catalog management to complex financial reporting.

Robust Architecture: The Django-based architecture provides security, scalability, and maintainability while following industry best practices.

Data Integrity: Through careful database design, transaction management, and signal-based updates, the system maintains data consistency across all modules.

User Experience: The intuitive interface and responsive design ensure accessibility for users with varying technical expertise.

Business Value: The system delivers measurable improvements in operational efficiency, accuracy, and decision-making capability.

The implementation demonstrates that modern web frameworks like Django, combined with thoughtful design and comprehensive testing, can deliver enterprise-grade solutions suitable for real-world business operations.

16.1 Future Enhancements

- Mobile Application Development
- Advanced Analytics and Reporting
- Integration Capabilities
- Advanced Features
- Performance Optimization
- Security Enhancements

16.2 Final Remarks

The Supermarket Electronic Management System represents a significant advancement in retail management technology, particularly for markets requiring multi-language and multi-calendar support. The successful implementation validates the chosen technology stack and architectural decisions.

The system's modular design and comprehensive API foundation provide a solid base for future enhancements. As businesses grow and requirements evolve, the system can be extended without fundamental architectural changes.

The project demonstrates that with careful planning, robust implementation, and thorough testing, it is possible to create complex enterprise systems that deliver real business value while maintaining code quality and maintainability.

This work contributes to the growing body of knowledge in enterprise software development and provides a practical example of Django framework capabilities in building production-grade business applications. The lessons learned and patterns established can guide future development efforts in similar domains.

17. References / Bibliography

1. Django Software Foundation. (2024). Django Documentation. <https://docs.djangoproject.com/>
2. PostgreSQL Global Development Group. (2024). PostgreSQL Documentation. <https://www.postgresql.org/docs/>
3. Django REST Framework. (2024). Django REST Framework Documentation. <https://www.django-rest-framework.org/>
4. Two Scoops Press. (2023). Two Scoops of Django 4.x: Best Practices for Django Web Development.
5. Mozilla Developer Network. (2024). Web Development Best Practices. <https://developer.mozilla.org/>
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